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Peer review form

Project reviewed: Reviewer: Date:

Your job is **not** to judge whether the model is good. It is to find the places where the conclusion might not hold. A review that finds a real flaw is more useful to your colleague than one that says the work is fine.

Be specific and be kind. Point at cells and lines.

1. Does it run?

- ☐ The notebook runs top to bottom in the course container without errors
- ☐ A second run produces the same numbers
- ☐ No package outside the course image is needed

If it failed, where?

2. Is the evaluation honest?

This is the heart of the review. Work through each question.

Was the evaluation designed before the modelling, or fitted around the result?

Could anything from the test set have influenced a decision? Check scaling, imputation, feature selection, encoding of categories, and hyperparameter tuning. Look specifically for a transformer fitted on the full dataset before splitting.

Does the splitting scheme suit the data? If there is a temporal ordering, a group structure, or repeated measurements of the same entity, a random split will overestimate performance. Is that handled?

Is the metric the right one for the problem? Accuracy on imbalanced data, R^2 on a skewed target, a single threshold with no justification — flag any of these.

Is the comparison between models fair? Same data, same preprocessing, same tuning effort for each?

3. Is the reasoning sound?

Is there a baseline, and is the improvement over it meaningful?

Do the stated conclusions follow from the evidence shown? Look for claims the numbers do not support.

Is the uncertainty acknowledged? A single number with no sense of its variability across folds or seeds is a weak result.

4. Can you follow it?

Could you reproduce this study from the write-up alone?

Which part was hardest to follow, and why?

5. The most useful thing you can say

One change that would most improve this project:

One thing done well that you intend to steal for your own:

Reviewer checklist

- ☐ I read the whole notebook, not only the conclusions
- ☐ I checked the preprocessing for leakage specifically
- ☐ My comments point at specific cells
- ☐ I said at least one concrete thing that would improve the work